

Roll No.

TCE-701

B. TECH. CIVIL ENGINEERING (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023

DESIGN OF STEEL STRUCTURES

Time : 1½ Hours

Maximum Marks : 50

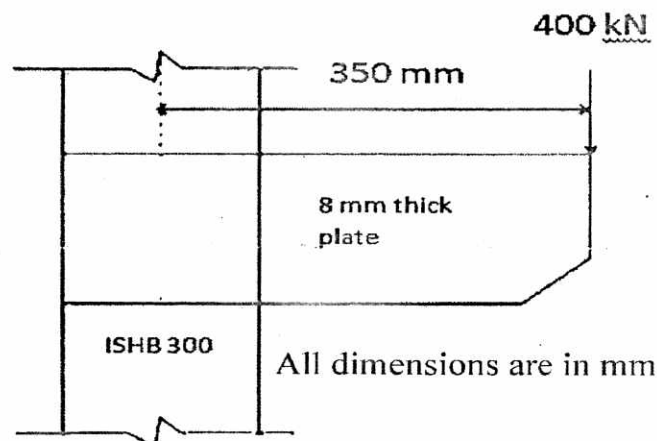
Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) With the help of a neat diagram, explain the engineering stress-strain curve for mild steel in detail. (CO1)

OR

- (b) A bracket is bolted to the flange of a column as shown below using 8 mm thick plate. Using M20 bolts of grade 4.6. Design the connection for $f_u = 410$ MPa, $f_{ub} = 400$ MPa, $f_y = 250$ MPa, $f_{yb} = 240$ MPa. (CO1)



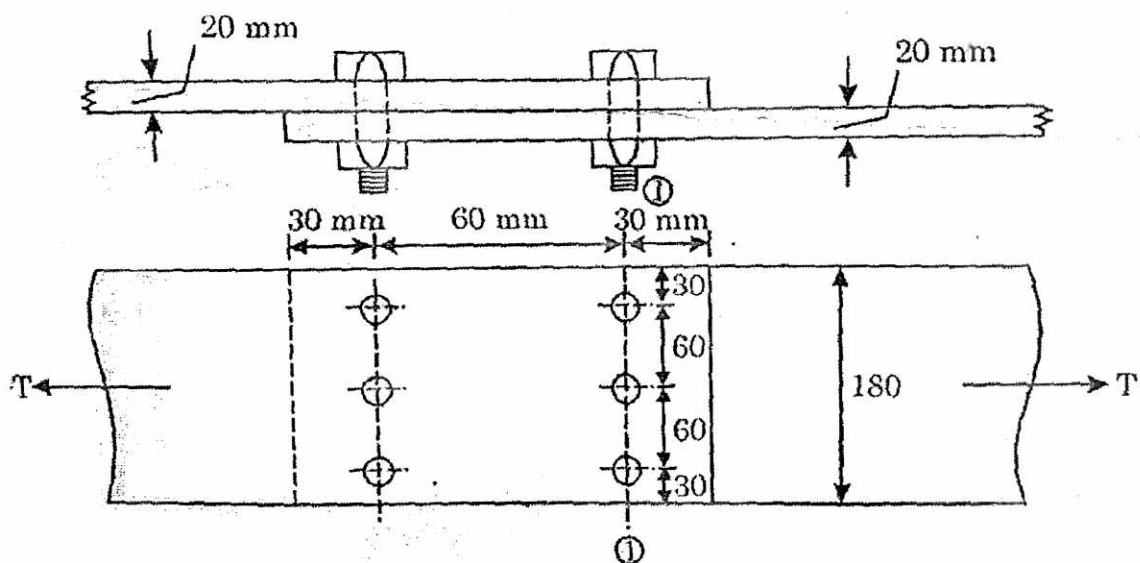
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2. (a) You have learnt about the tensile strength of mild steel, so can you explain the engineering stress strain curve for mild steel in detail. (CO1)

OR

- (b) Find the efficiency of the lap joint shown below.

Given : M 20 bolts of grade 4.6 are used and plate is Fe 410 with f_y 250 Mpa. (Shear plane cuts the thread). (CO1)

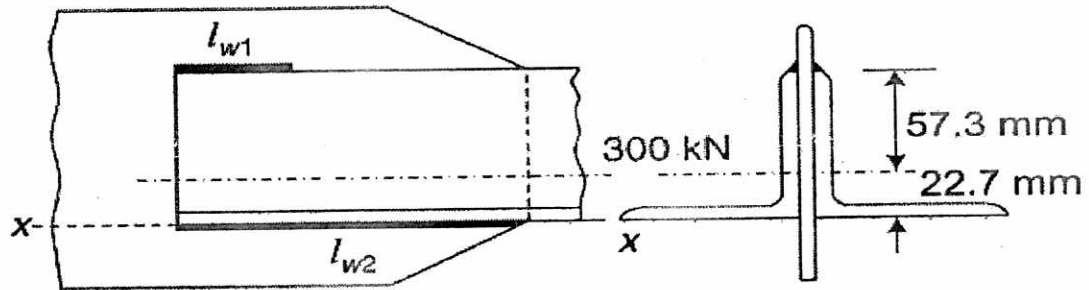


3. (a) Two flats (Fe 410 Grade Steel), each 210 mm $\times$ 8 mm, are to be jointed using 20 mm diameter, 4.6 grade bolts, to form a lap joint. The joint is supposed to transfer a factored load of 250 kN. Design the joint and determine suitable pitch for the bolts. (CO1)

OR

- (b) A tie member of a truss consists of double angle section, each 80 mm $\times$ 80 mm $\times$ 8 mm welded on the opposite of a 12 mm thick gusset plate as shown in Fig. Design a fillet weld for making the connection in the

workshop. The factored tensile force in the member is 300 kN. (Fe 410 grade steel for both connection and members). (CO1)



4. (a) Draw and explain various types of welding connections. Explain size of weld, throat of weld, and length of weld. (CO2)

OR

- (b) A groove weld is to connect two plates 180 mm $\times$ 18 mm each. Determine the design bending strength of the joint, if it is subjected to a moment of 13 kN-m. Also, determine the adequacy of the joint if the shear force at the joint is 200 kN. Assume the welds to be of double-U shop welded. (CO2)
5. (a) Design a bridge truss diagonal subjected to a factored tensile load of 300 kN. The length of the diagonal is 3.0 m. The tension member is connected to a gusset plate 16 mm thick with one line of 20 mm diameter bolts of grade 8.8. (CO2)

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(4)

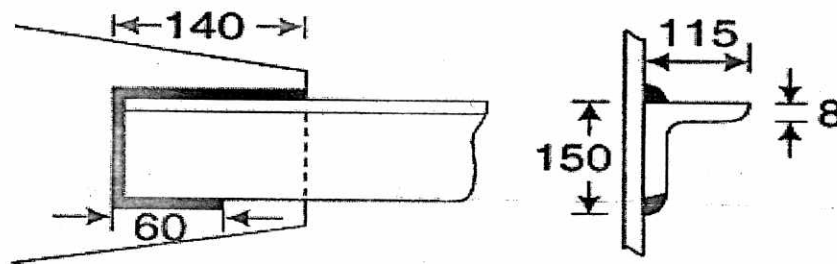
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OR

- (b) Compute the tensile strength of an section ISA $150 \times 115 \times 8$ mm of Fe 410 grade of steel connected with the gusset plate as shown in Fig. for the following cases : (CO2)

(a) Gross section yielding

(b) Net section rupture



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TCE-702

B. TECH. (CIVIL ENGINEERING) (SEVENTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

CONSTRUCTION MANAGEMENT AND PLANNING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Describe capital investment proposals, net present value with suitable example. (CO1)

OR

- (b) Explain in detail main causes of project failure. (CO1)

2. (a) Explain the importance of financial evaluation of projects in civil construction projects works. (CO1)

OR

- (b) Explain the role and importance of project planning in any project.

(CO1)

P. T. O.

(2)

3. (a) Describe the role of project development process in project management. (CO2)

OR

- (b) Describe Categories of construction projects in civil engineering works. (CO2)

4. (a) Explain the role of project management organization in civil construction projects. (CO2)

OR

- (b) Define staffing and also describe the importance of staffing in project management. (CO2)

5. (a) Describe the various stages and steps involved in project planning and also explain the importance of steps in project planning. (CO2)

OR

- (b) Explain the plan development process and its effect on project management. (CO2)

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TCE-703

B. TECH. (CIVIL) (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023
TRANSPORTATION ENGINEERING—II

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) Define gauges of a railway track. Enumerate different gauges used in India. (CO1)

OR

- (b) Discuss in brief the various traction on railway. (CO1)

2. (a) What are the possible causes of creep ? What are the effects of creep ? (CO1)

OR

- (b) Discuss in the brief the hauling capacity of locomotive. (CO1)

3. (a) Describe the function and requirements of rails in railway track. (CO2)

OR

- (b) What is the ballast in permanent way ? Mention the function of the ballast. (CO2)

P. T. O.

(2)

4. (a) What is the necessity of welding the rails ? How is the welding rails carried out ? (CO2)

OR

- (b) Define the criteria for deciding the length of rail. (CO2)

5. (a) Discuss the necessity and effects of coning of wheel. (CO2)

OR

- (b) Explain the different types of sleepers in use in Indian railways and state relative merit and demerit of each. (CO2)

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TCE-712

B. TECH. (CIVIL ENGINEERING) (SEVENTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

GEOGRAPHIC INFORMATION SYSTEMS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Describe the essential components of a GIS system in detail. Discuss the hardware and software requirements necessary for the effective functioning of a GIS. How do these components work together to process spatial data ? (CO1)

OR

- (b) Discuss the concept of topology in Geographic Information Systems (GIS). Explain its importance in spatial data representation and analysis. Provide examples to illustrate how topological relationships are used in GIS applications. (CO1)
2. (a) Describe the essential components of a GIS system in detail. Discuss the hardware and software requirements necessary for the effective functioning of a GIS. How do these components work together to process spatial data ? (CO1)

P. T. O.

OR

- (b) Explain the differences between the raster data model and the vector data model in GIS. Discuss their respective advantages and disadvantages, and provide real-world scenarios where each model is more suitable for representing spatial data. (CO1)
3. (a) Data acquisition is a critical aspect of GIS. Compare and contrast the methods of data acquisition, specifically focusing on scanners and digitizers. Provide examples of scenarios where each method would be preferred. (CO2)

OR

- (b) Describe various data compression techniques used in GIS, including Run Length Coding and Quadtree Tessellation. Provide a detailed explanation of each technique, its applications, and its impact on data storage and processing. (CO2)
4. (a) Explain the process of data storage, verification, and editing in GIS. Why are data verification and editing essential to maintaining data accuracy? Provide practical examples of data verification and editing tasks. (CO2)

OR

- (b) Discuss the critical aspects of map projection and transformation in GIS. Explain why map projection is necessary and how it influences spatial data accuracy. Provide examples of commonly used map projections. (CO2)